

JEE MAIN REVISION MASTER: HEAT AND THERMODYNAMICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 28 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	First Law of Thermodynamics & PV Thermodynamic Processes	18 PYQs	Thermodynamic Laws
High	Carnot Heat Engine Efficiency & Refrigerator COP	10 PYQs	Heat Engines

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
First Law of Thermodynamics & PV Thermodynamic Processes	$\Delta Q = \Delta U + \Delta W$, $PV^\gamma = \text{const}$, $W_{\text{iso}} = nRT \ln \frac{V_2}{V_1}$, $\Delta U = n C_v \Delta T$
Carnot Heat Engine Efficiency & Refrigerator COP	$\eta = 1 - \frac{T_2}{T_1} = \frac{Q_1 - Q_2}{Q_1}$, $\beta = \frac{T_2}{T_1 - T_2}$, $T(\text{K}) = T(^{\circ}\text{C}) + 273$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 First Law of Thermodynamics & PV Thermodynamic Processes (Very High Yield — 18 PYQs)

Core Principle: First Law: $\Delta Q = \Delta U + \Delta W$. In isothermal process ($\Delta T = 0$), $\Delta U = 0 \implies Q = W = nRT \ln(V_2/V_1)$. In adiabatic process ($Q = 0$), $PV^\gamma = \text{const}$ and $W = -\Delta U = (nR(T_1 - T_2) / (\gamma - 1))$. In isochoric process ($\Delta V = 0$), $W = 0$. In isobaric process, $W = P \Delta V$.

■ **Caution / JEE Pitfall:** Slope of an adiabatic curve on a P-V diagram is γ times steeper than the isothermal curve: $((dP/dV))_{\text{adia}} = \gamma ((dP/dV))_{\text{iso}}$.

■ **Speed Trick:** Work done in a cyclic process equals the enclosed area of the loop on the P-V diagram. Clockwise cycle $\implies W > 0$ (Heat engine); Counter-clockwise $\implies W < 0$ (Refrigerator).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Following are statements of a few processes taking place in nature. I. Free expansion of a gas II. The combustion of a mixture of petrol and..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "In thermodynamic processes, which of the following statements is not true?..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "The graph of pressure P and volume V of 1 mole of an ideal gas at constant temperature is..." [Ans: N/A]

3.2 Carnot Heat Engine Efficiency & Refrigerator COP (High Yield — 10 PYQs)

Core Principle: Carnot engine efficiency $\eta = 1 - (T_2 / T_1) = (W / Q_1)$ (temperatures strictly in Kelvin). Coefficient of performance (COP) of Carnot refrigerator $\beta = (Q_2 / W) = (T_2 / T_1 - T_2)$. Relation between β and η : $\beta = (1 - \eta) / \eta$.

■ **Caution / JEE Pitfall:** Substituting Celsius instead of Kelvin temperatures into $\eta = 1 - T_2/T_1$. ALWAYS convert to Kelvin first!

■ **Speed Trick:** To double the efficiency of a Carnot engine, it is always more effective to lower the sink temperature T_2 than to raise the source temperature T_1 by the same amount.

Representative Source PYQs:

- **[2025 Shift] Q#2 (MCQ):** "At 27°C temperature, the mean kinetic energy of the atoms of an ideal gas is E_1 . If the temperature is increased to 327°C ..." **[Ans: N/A]**
- **[2025 Shift] Q#4 (MCQ):** "Three metal rods of the same material and identical in all respects are joined as shown in the figure. The temperatures at the ends of these..." **[Ans: N/A]**
- **[2024 Shift] Q#4 (MCQ):** "A solid cube of mass m at a temperature θ_1 is heated at a constant rate. It becomes liquid at temperature θ_2 and vapour..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Following are statements of a few processes taking place in nature. I. Free expansion of a gas II. The combustion of a mixture of petrol and air ignited by a spark III. The leaking of gas from the kitchen cylinder. IV. The transfer of heat from one heated part of a liquid to the other colder part. Which amongst these processes are irreversible in nature?

- (A) I and II
- (B) I and III
- (C) II, III and IV
- (D) I, II, III and IV

Correct Answer: None

Solution Analysis: All the given processes are irreversible in nature. Free expansion of a gas is irreversible because the gas spreads into the available space on its own. It cannot return to its original volume without some external work being done. Combustion of petrol and air is also irreversible. Once petrol burns, it forms new substances like carbon dioxide, water vapour, etc. These products cannot change back into petrol and air by simply reversing the process. Leaking of gas from a kitchen cylinder is irreversible because the gas moves from high pressure inside the cylinder to lower pressure outside. It w

Benchmark Problem 2: KCET 2025 (Q#2)

At 27°C temperature, the mean kinetic energy of the atoms of an ideal gas is E_1 . If the temperature is increased to 327°C, then the mean kinetic energy of the atoms will be

- (A) $\frac{E_1}{\sqrt{2}}$
- (B) $\sqrt{2}E_1$
- (C) $2E_1$
- (D) $\frac{E_1}{2}$

Correct Answer: None

Solution Analysis: The mean kinetic energy of the atoms in an ideal gas is directly proportional to the temperature measured in Kelvin. At an initial temperature of 27°C, the mean kinetic energy is E_1 . When the temperature is increased to 327°C, we can determine the new mean kinetic energy E_2 using the relationship:

$\frac{E_1}{E_2} = \frac{T_1}{T_2}$ To find E_2 , we can rearrange this equation: $E_2 = E_1 \frac{T_2}{T_1}$ Converting the given temperatures from Celsius to Kelvin: Initial temperature, $T_1 = 27^\circ\text{C} + 273 = 300\text{K}$ Final temperature, $T_2 = 327^\circ\text{C} + 273 = 600\text{K}$ Substitu